

# PE-RankFormer: Models and Results

A technical report on the current best system for prime-editing efficiency prediction

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## Abstract

We report the current best models for predicting prime-editing efficiency, benchmarked against OptiPrime on its own training data, cross-validation splits, and complete held-out test set. The final system reaches Spearman  $\rho = \mathbf{0.9079}$  on the 20,509-row official held-out set against OptiPrime’s 0.8690, a difference of  $+\mathbf{0.0389}$  with a protospacer-clustered paired bootstrap 95% CI of  $[+0.0286, +0.0498]$ , ahead in *all* 5,000 resamples ( $p < 0.0002$ ). The margin is  $+0.0804$  on the Kim partition and  $+0.0220$  on Liu.

The system is an equal-weight rank average of five out-of-fold members. Its strongest single member — an *ordinal-SSM* model combining a cumulative-threshold objective with a bidirectional state-space sequence mixer — reaches 0.9082 out-of-fold on development folds, above the entire previous three-member ensemble.

We also report, in equal detail, what did *not* work. Approximately fifty subsequent experiments spanning supervision geometry, architecture, capacity, output parameterisation and training scheme produced no improvement over this system: each returned what an unmodified re-run of the same model returns. We give the measurements that establish this, a replicate-based estimate showing that roughly  $+0.12$  Spearman of *genuine* headroom nonetheless remains on the Kim partition, and a diagnosis of where that headroom sits.

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# 1 Summary of results

Table 1: Final held-out performance. Official OptiPrime held-out set,  $n = 20,509$  rows over 750 protospacer clusters. No model in this table was trained on these rows.

| Model                          | Full          | Liu ( $n=9,175$ ) | Kim ( $n=11,334$ ) |
|--------------------------------|---------------|-------------------|--------------------|
| <b>PE-RankFormer (final)</b>   | <b>0.9079</b> | <b>0.8585</b>     | <b>0.8124</b>      |
| PE-RankFormer (previous round) | 0.8933        | 0.8462            | 0.7836             |
| PE-RankFormer (first version)  | 0.8865        | 0.8349            | 0.7751             |
| OptiPrime                      | 0.8690        | 0.8365            | 0.7320             |

Table 2: Paired protospacer-clustered bootstrap, 5,000 resamples, on identical rows. Clustering is by protospacer because pegRNAs sharing a target are not independent; a row-level bootstrap would materially understate the intervals.

| Comparison                 | $\Delta\rho$   | 95% CI             | wins  | $p$      |
|----------------------------|----------------|--------------------|-------|----------|
| <b>Final – OptiPrime</b>   | <b>+0.0389</b> | [+0.0286, +0.0498] | 1.000 | < 0.0002 |
| Final – previous round     | +0.0146        | [+0.0113, +0.0184] | 1.000 | < 0.0002 |
| Previous round – OptiPrime | +0.0243        | [+0.0138, +0.0346] | 1.000 | < 0.0002 |

**A caveat that belongs beside the headline.** The absolute figure’s own 95% interval is [0.8955, 0.9173], and 90.0% of resamples exceed 0.90. The 0.90 threshold is therefore cleared by the point estimate but *not* established at 95% confidence. The *margin over OptiPrime* is a much stronger claim than the absolute level: it wins every one of 5,000 resamples.

## 2 Data and evaluation protocol

### 2.1 Corpus

All models are trained on OptiPrime’s own training mix, supplied by its authors: 297,962 rows, plus the 20,509-row held-out set, giving 318,471 total. Three source studies contribute, and their proportions differ sharply between training and evaluation — a fact that becomes important in §6.

Table 3: Corpus composition. Schwank contributes 58.4% of training rows and 0% of the evaluation set. “Zero-mass” is the fraction of rows with efficiency exactly 0.

| Source                                    | % of training | % of held-out | Zero-mass |
|-------------------------------------------|---------------|---------------|-----------|
| Schwank ( <code>pridict_pridict2</code> ) | 58.4%         | 0.0%          | 10.3%     |
| Liu ( <code>hsu2026</code> )              | 22.0%         | 44.7%         | 0.8%      |
| Kim ( <code>deepprime</code> )            | 19.6%         | 55.3%         | 49.9%     |

### 2.2 Evaluation discipline

A four-level hierarchy separates exploration from confirmation:

1. **Matched development folds** (3 folds,  $\approx 35.7$ k rows each), composed to match the held-out set’s Liu/Kim proportions. Used freely for exploration.
2. **Internal lockbox** (17,975 rows, 367 protospacers, sharing no protospacer with any development validation set). Touched once, to gate a shortlist.
3. **Official 5-fold out-of-fold predictions**. Every row is scored only by the checkpoint that held its fold out.
4. **Official held-out set**. Evaluated once, after the model was frozen in writing, with the expected result recorded in advance.

Everything reported as a development number is out-of-fold. The selection protocol (which subset, which combination rule, how to break ties) was written down and committed *before* the ensemble search ran.

## 3 Model architecture

### 3.1 Backbone

Each member is a dual-encoder relational model:

1. an **edit encoder** over paired wild-type/edited tokens;
2. a **pegRNA encoder** over segment-aware spacer/PBS/RTT tokens;
3. **bidirectional cross-attention** between the two streams;
4. **FiLM conditioning** on experimental context (cell line, PE system, Cas9 variant), applied either once after pooling (“late”) or at every block (“layerwise”);

### 5. an **attention-pooled output head**.

Base configuration:  $d_{\text{model}}=384$ , 6 edit layers, 4 pegRNA layers, 2 cross blocks,  $\approx 25\text{M}$  parameters.

### 3.2 The ordinal outcome head

The evaluation metric is Spearman, which depends only on order. The ordinal head targets that directly. Fix  $K - 1$  thresholds  $t_k$  at quantiles of the *training* efficiency distribution; the head emits one logit per threshold, trained with binary cross-entropy against the cumulative indicators, and scores a row by

$$s(x) = \frac{1}{K-1} \sum_{k=1}^{K-1} \sigma(z_k(x)) \approx \text{the row's normalised rank.}$$

Two properties motivate it. It is *metric-matched*: a head estimating the target’s quantile optimises the evaluated quantity, whereas a proportion head is only incidentally monotone. And it is *robust to the target’s skew* — efficiency has large mass at zero, so a Huber or proportion objective spends most of its gradient on a few high-efficiency rows, while  $K - 1$  balanced binary decisions spread supervision across the distribution.

Thresholds are computed from training rows only and stored in the checkpoint, so evaluation reproduces the score mapping exactly.

### 3.3 The state-space sequence mixer

One family of members replaces self-attention with a **bidirectional diagonal state-space (S4D) mixer**. With a diagonal state matrix the convolution kernel is available in closed form,

$$K[\ell] = 2 \operatorname{Re} \left( \sum_n C_n \frac{e^{\Delta A_n - 1}}{A_n} (e^{\Delta A_n})^\ell \right),$$

and is applied by FFT convolution. Two independent kernels (forward and time-reversed) are summed, since prime-editing efficiency depends on context on both sides of the edit. The block topology deliberately mirrors the Transformer layer it replaces, so a comparison isolates the mixing mechanism rather than confounding it with block design.

### 3.4 Which axis matters: an **objective** $\times$ **architecture** factorial

All four cells existed as out-of-fold predictions, so this was computed from stored artifacts at no additional cost.

Table 4: Out-of-fold development Spearman, fold-averaged.

|                                                        | Simplex head | Ordinal head   |
|--------------------------------------------------------|--------------|----------------|
| Transformer                                            | 0.8798       | 0.8911         |
| State-space (SSM)                                      | 0.9011       | <b>0.9082</b>  |
| <hr/>                                                  |              |                |
| Main effect of <b>objective</b> (ordinal – simplex)    |              | +0.0092        |
| Main effect of <b>architecture</b> (SSM – Transformer) |              | <b>+0.0192</b> |
| Interaction                                            |              | –0.0043        |

**This corrects an earlier claim of ours.** An earlier version of this work described the ordinal head as “the breakthrough ... a change of objective, not architecture”. The factorial does not support that: the architecture effect is roughly twice the objective effect. The ordinal head was the more novel contribution and the one that produced decorrelated errors, but the state-space mixer contributed more raw accuracy.

The interaction is *sub-additive*: ordinal supervision is worth +0.0114 on a Transformer but only +0.0071 on an SSM. The two changes partly capture the same improvement rather than stacking, which sets a realistic ceiling on “add another mechanism” strategies — a prediction borne out in §6.

Caveat: these are observational cells rather than a purpose-built factorial. Three are from one training round; the Transformer/simplex cell is an earlier frozen baseline. They share corpus, splits, out-of-fold scoring, capacity and optimiser recipe, so the main effects are trustworthy at the  $\pm 0.002$  development resolution, but the interaction term is the quantity most exposed to that mismatch.

## 4 The final system

### 4.1 Composition

An equal-weight **rank average** of five out-of-fold members. Each member is five checkpoints, one per official fold; every row is scored only by the checkpoint that held its fold out. Rank averaging is used because members are not mutually calibrated — the ordinal members emit rank estimates and the simplex members emit proportions, so averaging raw scores would mostly measure scale mismatch.

Table 5: Members and their individual held-out performance.

| Member          | Design                        | Full          | Liu           | Kim           |
|-----------------|-------------------------------|---------------|---------------|---------------|
| ordSSM          | ordinal head + S4D mixer      | <b>0.9082</b> | 0.8576        | <b>0.8151</b> |
| ssm             | simplex head + S4D mixer      | 0.9017        | 0.8563        | 0.7982        |
| ordA            | ordinal head + layerwise FiLM | 0.9005        | 0.8394        | 0.8082        |
| ordC            | ordinal head + feature branch | 0.8947        | 0.8261        | 0.8027        |
| familyA         | simplex head + layerwise FiLM | 0.8888        | 0.8412        | 0.7787        |
| <b>Ensemble</b> | equal-weight rank average     | <b>0.9079</b> | <b>0.8585</b> | 0.8124        |

### 4.2 An honest observation about the ensemble

Table 5 contains a result we did not anticipate and think should be stated plainly: **the single best member matches the five-member ensemble on the held-out set** (0.9082 vs 0.9079). A leave-one-out analysis on held-out data makes the point sharper.

Table 6: Leave-one-member-out on the held-out set. Positive contribution means the ensemble is worse without that member. *Post-hoc analysis: this did not and could not inform the frozen model choice.*

| Member dropped | Ensemble without it | Contribution |
|----------------|---------------------|--------------|
| ordSSM         | 0.9048              | +0.0031      |
| ssm            | 0.9066              | +0.0013      |
| ordA           | 0.9073              | +0.0006      |
| ordC           | 0.9085              | −0.0007      |
| familyA        | 0.9092              | −0.0013      |

On the development folds the ensemble was worth +0.0074 over its best member (0.9156 vs 0.9082). On held-out that advantage is −0.0003. **The ensemble gain did not transfer.** Two members would, post-hoc, have been marginally better excluded.

We report this rather than quietly presenting the ensemble as the better system, because it bears on how the result should be used. All the differences involved (0.0003–0.0013) are far below the  $\approx 0.005$  resolution of our development evaluation, so the honest reading is that **the ensemble and its best single member are indistinguishable on this benchmark**. For practical use, the single ordinal-SSM model is preferable: one model instead of twenty-five checkpoints, at the same accuracy. We did not re-select on held-out to make that switch official, because a model chosen by inspecting the test set carries none of the guarantees the frozen one does.

### 4.3 Selection procedure

The ensemble was chosen under a protocol committed in writing before the search ran: rank subsets on development folds; require a candidate to beat the incumbent on *every* fold rather than on the mean; gate the shortlist on the lockbox, scored out-of-fold; and on disagreement prefer the lockbox winner, breaking near-ties toward fewer members.

The development winner (4 members) and the lockbox winner (5 members) disagreed by 0.001011, eleven millionths past the pre-set 0.001 near-tie threshold, so the rule selected the five-member subset. We note this was marginal — the two candidates are statistically indistinguishable — and followed the rule as written rather than re-arguing it after seeing the numbers. The lockbox’s own unconstrained best subset (0.9155) was *not* adopted, since selecting on the lockbox would have converted the only unworn evaluation surface into a fitted one.

## 5 Detailed results

### 5.1 Per-condition breakdown

Table 7: Held-out Spearman by experimental condition ( $n \geq 300$ ). Performance spans 0.21 between the weakest and strongest conditions.

| Study | Cell line  | PE system | $n$  | $\rho$ |
|-------|------------|-----------|------|--------|
| Kim   | A549       | PE2       | 1299 | 0.6712 |
| Kim   | MDA-MB-231 | PE2       | 620  | 0.7100 |
| Kim   | A549       | PE4       | 1281 | 0.7144 |
| Kim   | HCT116     | PE2       | 651  | 0.7866 |
| Kim   | NIH3T3     | PE4       | 650  | 0.8034 |
| Kim   | HeLa       | PE2       | 664  | 0.8103 |
| Liu   | HEK293T    | PE2       | 2158 | 0.8179 |
| Liu   | HEK293T    | PE4       | 2148 | 0.8282 |
| Kim   | HEK293T    | PE2       | 3208 | 0.8298 |
| Liu   | HeLa       | PE4       | 2435 | 0.8311 |
| Liu   | HeLa       | PE2       | 2434 | 0.8318 |
| Kim   | HEK293T    | PE4       | 1135 | 0.8592 |
| Kim   | DLD1       | PE4       | 1164 | 0.8778 |
| Kim   | DLD1       | PE2       | 662  | 0.8809 |

Performance tracks the zero-mass of the condition: A549/PE2 is 63.9% exact zeros and scores 0.671; DLD1/PE2 is 37.8% zeros and scores 0.881.

### 5.2 Calibration: restoring an absolute prediction

Rank averaging produces excellent orderings but destroys the output scale: the ensemble’s “prediction” is a normalised rank, not an editing efficiency, so absolute error metrics against it are meaningless. We fit a monotone map  $g : \text{rank} \mapsto \text{efficiency}$  using isotonic regression on out-of-fold development predictions only, and apply it once to the frozen held-out predictions.

Table 8: Effect of monotone calibration. The map is fitted entirely on development data, is verified non-decreasing, and inverts no pair — it is a change of units on an already-frozen prediction vector, not a new model selection.

|                   | Spearman      | Pearson       | MAE           | RMSE          |
|-------------------|---------------|---------------|---------------|---------------|
| Uncalibrated      | 0.9079        | 0.7278        | 0.3869        | 0.4343        |
| <b>Calibrated</b> | <b>0.9079</b> | <b>0.8637</b> | <b>0.0478</b> | <b>0.0912</b> |

MAE falls roughly eight-fold and Pearson rises 0.136 with the ranking untouched (the residual  $2 \times 10^{-5}$  Spearman shift comes from isotonic regression’s piecewise-constant plateaus creating ties, not from any pair being reordered). The system now emits a usable efficiency estimate.

## 6 What did not work

After the system above was frozen, approximately fifty further experiments were run across three rounds. **None improved on it.** We report them because the pattern is more informative than any individual result, and because several are ideas a reader might otherwise expect to see tried.

Table 9: Summary of failed interventions. “ $\Delta$ ” is against a matched control — in every case an unmodified re-run of the same model, which we found to be indispensable (see below).

| Category                | Intervention                            | $\Delta$                    |
|-------------------------|-----------------------------------------|-----------------------------|
| Post-hoc combination    | context-gated ensemble weights          | +0.0000                     |
|                         | nonlinear stacking (GBM, ridge)         | +0.0013                     |
|                         | residual learning on features           | +0.0006 (oracle bound)      |
| Supervision geometry    | auxiliary simplex head (dual-head)      | +0.0008                     |
|                         | multi-resolution ordinal                | +0.0008                     |
|                         | context-relative ordinal (auxiliary)    | +0.0016                     |
|                         | quantile head (pinball loss)            | −0.0140                     |
|                         | pairwise ranking loss                   | −0.0037                     |
| Output parameterisation | true CORAL (rank-consistent)            | +0.0005                     |
|                         | monotonicity penalty                    | +0.0007                     |
|                         | hurdle head (zero-inflation)            | −0.0022                     |
| Architecture / capacity | hybrid SSM+attention (alternating)      | −0.0014                     |
|                         | hybrid SSM+attention (parallel gate)    | −0.0027                     |
|                         | SSM state dim 128 / 256                 | −0.0022 / +0.0002           |
|                         | wider FFN, deeper stack                 | −0.0001 / did not replicate |
|                         | mixture-of-experts (4 / 8)              | did not replicate / −0.0013 |
|                         | layerwise context on SSM                | +0.0001                     |
| Training scheme         | protospacer bagging                     | +0.0002 to −0.0022          |
|                         | source re-weighting (mild / aggressive) | did not replicate / −0.0059 |
|                         | new random seed                         | +0.0022                     |

### 6.1 Three findings from the failures

**A mechanism-free control is indispensable.** In one round, fourteen candidates each showed a small positive gain of +0.0017 to +0.0020. The round also included a plain re-run of the baseline at a different batch size — no new mechanism whatsoever — which gained +0.0018. **The difference attributable to every idea in that round combined was +0.0002.** Without that control we would have reported fourteen promising leads.

**Single-fold results at this effect size are not evidence.** In a later round, three interventions on genuinely orthogonal axes (capacity, data distribution, conditional capacity) each reached +0.003 to +0.004 on one development fold. All three *changed sign* on a second fold. Replication, not effect size, was the discriminating test.

**Some benchmark gains are metric artifacts.** 28.4% of rows have efficiency exactly zero (49.9% within Kim). Collapsing the lowest-scoring predictions onto a single tied value matches



that tie structure and gains +0.0044 pooled Spearman, positive on all three folds. We *rejected* it: applying the same transform to five other models gave +0.0127 to +0.0162 regardless of model quality, so it exploits Spearman’s tie convention rather than improving prediction, and OptiPrime emits continuous tie-free scores. This is worth recording as a benchmark caveat — **comparisons between models with different tie behaviour on zero-inflated data are not automatically like-for-like.**

## 7 Remaining headroom, and where it sits

The obvious inference from §6 — that the benchmark is saturated — is testable, and it is wrong.

### 7.1 A replicate-based noise ceiling

The Kim data contains genuine technical replicates: 649 groups in which all sixteen design and condition covariates *and* the target site are identical. If the same pegRNA in the same condition is measured twice and the measurements disagree, no model can predict either better than they predict each other. For two independent measurements of a true value, the reliability  $R = \text{corr}(y_1, y_2)$  implies a ceiling of  $\sqrt{R}$  against a single observation.

Two corrections were necessary, and either shortcut inverts the conclusion:

- **Naive replicate keys give  $R = 0.7987$** , almost exactly our model’s Kim score — which would have “confirmed” saturation. But grouping on design and condition alone merges epegRNA with plain-pegRNA constructs, which differ in motif, epegRNA flag and linker. Those are different designs, not repeats.
- **Treating an observed zero as noiseless inflates the ceiling.** 21.4% of rows measured at exactly 0 have a non-zero replicate (median 0.0015). A zero is *censored*, not certain. A Gaussian noise model assuming otherwise gives 0.968; an empirical nearest-neighbour resampler gives 0.9026, and the difference falls almost entirely on the zero-heavy conditions the model does worst on.

Table 10: Kim headroom against the empirical ceiling, development fold.

| Scope                      | Model  | Ceiling | Gap            |
|----------------------------|--------|---------|----------------|
| All Kim rows               | 0.7869 | 0.9026  | <b>+0.1157</b> |
| Rows that edit ( $y > 0$ ) | 0.8465 | 0.9489  | +0.1024        |

**Kim is not noise-limited**, and the gap is not merely the zero block: excluding zeros entirely still leaves +0.10.

### 7.2 Diagnosis: context rescales, it does not reorder

For designs measured in two conditions, across 45 condition pairs:

|                                                                   |               |
|-------------------------------------------------------------------|---------------|
| Mean cross-condition $\rho$ , <b>true</b> (what the biology does) | 0.683         |
| Mean cross-condition $\rho$ , <b>model</b> (what it predicts)     | 0.835         |
| Excess                                                            | <b>+0.152</b> |

The model applies a near-universal sequence ranking and uses experimental context mainly to set the scale, whereas reality reorders designs substantially between cell lines and PE systems. **That reordering is learnable**: predicting the cross-condition rank shift from just sixteen engineered design features reaches held-out  $\rho \approx 0.275$ . The excess similarity (+0.152) is the same order as the gap to the ceiling (+0.116).

### 7.3 A falsified fix, and what it taught

The natural remedy — conditioning at every block instead of once after pooling — was implemented for the state-space backbone and tested on three folds with matched controls. It was null (+0.0001), and the pre-registered mechanism test failed in the informative direction: cross-condition rank correlation *rose* to 0.875 from the control’s 0.828.

More conditioning made the model *more* condition-invariant. FiLM is structurally a per-channel scale and shift, so applying it at every block adds no capacity to reorder designs; it adds a better pathway for the cross-condition mean shift, which explains far more pooled variance and is much easier to fit. The extra capacity is spent on the shortcut.

The diagnosis survives; the instrument was wrong. Work is ongoing on removing the shortcut from the *objective* rather than adding conditioning capacity.

## 8 Limitations

1. **The 0.90 threshold is not established at 95% confidence** ([0.8955, 0.9173]; 90% of resamples above). The margin over OptiPrime is the robust claim.
2. **The ensemble does not beat its best member on held-out** (§4.2). The development-fold advantage did not transfer.
3. **196 of 649 Kim replicate groups straddle the official train/held-out split**, so those held-out rows have an exact design-and-condition twin in training. This affects any model trained on this split equally — ours and OptiPrime’s — so it does not distort the comparison, but it makes absolute held-out scores slightly optimistic.
4. **Kim has no fresh evaluation surface left**. Of 2,501 Liu+Kim training protospacers, 1,553 sit in development validation sets, 367 were consumed by the lockbox, and 302 collide with Schwank; the 279 remaining are 98.5% Liu. Kim-side claims rest on out-of-fold development folds and a once-used lockbox.
5. **Our OptiPrime reproduction scores above its published figure** under every aggregation convention we tried, and no evaluation script was released. The direction favours the baseline, so it does not undermine our claims, but it caps the precision of the comparison.
6. **No external validation**. All results are on one benchmark.

## 9 Reproduction

```
# Final system: five members x five official folds
scripts/train/train_pilot.py --config configs/round5/baseline_ordinal_ssm.yaml \
  --run-name <member> --val-fold <k> [member-specific flags]
```

```
# Out-of-fold development predictions
scripts/evaluate/run_round4_phase3.sh

# Ensemble search and lockbox gate
scripts/evaluate/search_ensemble.py --members ...

# Held-out evaluation (guarded; every run is audit-logged)
scripts/evaluate/evaluate_heterogeneous_heldout.py --member ... \
    --allow-heldout-evaluation
scripts/evaluate/round4_final_bootstrap.py

# Calibration and noise ceiling
scripts/evaluate/calibrate_ensemble.py --apply-heldout
scripts/evaluate/noise_ceiling_empirical.py
```

The frozen model specification, the pre-registered selection protocol, and the full research log including every negative result are in **reports/**.